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Evolution of shrinkage, real density, porosity, heat and mass transfer coefficients during indirect solar drying of cocoa beans

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ABSTRACT

Drying is a main process of conservation used for cocoa beans and it is so essential to control its drying parameters. In this study, shrinkage, density, porosity, and heat and mass transfer coefficients of cocoa beans during indirect solar drying were investigated. The results showed that shrinkage and porosity increased with decrease in reduced moisture content. The real density varied during drying process. Its value decreased from 825.10 kg/m³ at the beginning of the drying to about 696.25 kg/m³ at the end of the drying process. The cocoa beans had a final porosity approaching 25% and most of its removed water during drying was replaced by gas. Also, the values of heat and mass transfer coefficients increased from 1.92×10^{-4} to 8.08×10^{-2} W/m² K and from 1.88×10^{-7} to 7.88×10^{-5} m/s, respectively, for an indirect solar drying of cocoa beans. The effective moisture diffusivity was influenced by shrinkage. Effective moisture diffusivities were calculated by Fick's diffusion law and their values varied from 5.49×10^{-10} to 4.26×10^{-10} m²/s. The cocoa beans thermophysical properties were obtained and fitted to nonlinear correlations, describing their behaviour as a function of moisture content.

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1. Introduction

The processing of cocoa beans begins with the harvesting of the fully matured cocoa fruits which are usually oval in shape and measure between 15 and 30 cm long. Inside the fruit, there are about 30–40 cocoa beans that are covered with a whitish layer of sweet mucilaginous pulps (Hii, 2004). Upon harvesting of cocoa fruits, fresh cocoa beans are fermented for 5–7 days inside a wooden box and dried immediately to the desirable moisture content at 7–8% dry basis for safe storage (Hii et al., 2009a,b).

Drying is a key process in many food industries and in many agricultural countries. Large quantities of food products are dried to improve shelf life, reduce packaging cost, lower shipping weights, enhance appearance, encapsulate original flavour and maintain nutritional value. The primary objective of drying was to remove moisture from the food so that bacteria, yeast and

mould cannot grow and spoil the food (Chou and Chua, 2001; Siqueira et al., 2012). However, drying of food produces great changes in their volume and surface area simultaneously with loss of moisture. Therefore, it is necessary to devote more attention to the shrinkage phenomenon and the prominent porosity changes (Nadhari et al., 2014). The shrinkage phenomenon affects in particular the diffusion coefficient of the product, which is one of the main parameters governing the drying process; it also has an influence on the drying rate (De Lima et al., 2002; Liu et al., 2012). Besides, various characteristics of the product depend on its density so that the knowledge of the density changes with the moisture content will be useful to characterize the behaviour of this product (De Lima et al., 2002; Talla et al., 2004; Dissa et al., 2010).

As porosity, density and shrinkage are critical factors, influencing mass and heat transfer mechanism as well as product quality parameters, it becomes important to include porosity, density and shrinkage while predicting drying process and for better optimization of process (Aprajeeeta et al., 2015). Heat transfer occurs inside the product by conduction and mass transfer takes place by diffusion in accordance with temperature and moisture concentration gradient, respectively (Srikiatden and Roberts, 2008). Studies on heat and mass transfer are crucial especially in dryer design to determine the effect of various drying parameters on efficiency, cost and degradation of important product quality attributes (Bennamoun and Belhamri, 2008; Perussello et al., 2012). Thereby,

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Nomenclature

A	exchange surface (m ²)
a	shrinkage coefficient
C_p	specific heat (J/kg K)
D	moisture diffusivity (m ² /s)
d	geometric diameter (m)
F_m	mass flux (kg/m ² s)
h	heat transfer coefficient (W/m ² K)
h_m	mass transfer coefficient (m/s)
k	thermal conductivity (W/m K)
Le	Lewis number
L_v	latent heat of vaporization (J/kg)
m	mass (kg)
Nu	Nusselt number
R	radius (m)
r	radial coordinate
S	shrinkage
Sh	Sherwood number
T	temperature (°C)
t	time (s)
V	volume (m ³)

X	moisture content (kg water/kg db)
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Greek symbols

α	thermal diffusivity (m ² /s)
β	compactness (m ⁻¹)
ρ	density (kg/m ³)
ε	porosity (%)

Subscripts

a	air
b	bulk
d	dry
eff	effective
eq	equilibrium
O	initial
P	product
r	reduced
w	water

several studies on the mass and heat transfer of cocoa beans drying had been carried out (Nganhou et al., 1992; Daud et al., 1998; Kyi et al., 2005; Garcia-Alamilla et al., 2007; Hii et al., 2009a,b; Paramo et al., 2010; Hii et al., 2013). But these studies have been performed without considering shrinkage except for study carried out by Hii et al. (2013). Besides, there is not enough works which are carried out on the drying kinetics of cocoa beans taking into account the shrinkage phenomenon, the porosity and the density changes.

Thus, the objective of this present work was to determine and predict shrinkage, porosity and density changes occurring along with simultaneous mass and heat transfer during the indirect solar drying of cocoa beans.

2. Theoretical analysis**2.1. Drying rates**

The drying rate at different instants of drying was estimated by derivation of the moisture content with respect to drying time. However, the convective exchanges, at all scales, ask to quantify the exchange surface (A) contained in the volume (V) in which transfers are characterized. The ratio between these two quantities constitutes the compactness (β):

$$\beta = \frac{A}{V_d} \quad (1)$$

where V_d is the volume of dry product.

This product compactness varies during the drying and that must be taken into account in the drying rates curves representation. Assuming that the shrinkage is isotropic, the drying rate or mass flux (F_m) corrected for shrinkage deduced from the mass loss can be stated as (Dissa et al., 2010):

$$F_m = -\left(\frac{\rho_d}{\beta}\right) \frac{dX}{dt} = -\left(\frac{m_d}{A}\right) \frac{dX}{dt} \quad (2)$$

where ρ_d is the bulk density of dry product and m_d its dry mass.

The drying rates (dX/dt) were calculated by approximation of the derivatives to finite differences based on the following Eqs. (3)–(5).

$$\frac{dX}{dt} = \left(\frac{X_1 - X_0}{t_1 - t_0}\right) \quad \text{at initial instant } t_0 \quad (3)$$

$$\frac{dX}{dt} = \left(\frac{X_n - X_{n-1}}{t_n - t_{n-1}}\right) \quad \text{at instant } t_n \text{ of the end of drying} \quad (4)$$

$$\frac{dX}{dt} = \frac{1}{2} \left(\frac{X_i - X_{i-1}}{t_i - t_{i-1}} + \frac{X_{i+1} - X_i}{t_{i+1} - t_i} \right) \quad \text{at each instant } t_i \text{ for } i \text{ between } 1 \text{ and } n-1 \quad (5)$$

2.2. Real density

Assume that the total mass of the product is constituted of dry mass of solid, water mass and air mass. If the mass of air is neglected, the total mass of the product is given as follows:

$$m_p = m_d + m_w \quad (6)$$

The total volume of the product is as follows:

$$V_p = V_d + V_w + V_a \quad (7)$$

The real density of product (ρ_p) is the ratio between the current total mass of the product and the volume occupied by the solid and the liquid matter only (Koç et al., 2008):

$$\rho_p = \frac{m_p}{V_d + V_w} \quad (8)$$

It can be noted that the product density assumes $V_a = 0$.

Furthermore, it can be written as follows:

$$V = \frac{m}{\rho} \quad (9)$$

$$X = \frac{m_w}{m_d} \quad (10)$$

The development of Eqs. (6)–(10) gives the following expression (Talla et al., 2004; Koç et al., 2008):

$$\rho_p = \frac{1 + X}{\frac{1}{\rho_s} + \frac{X}{\rho_w}} \quad (11)$$

2.3. Shrinkage

The particular structure of the food materials and the mechanical characteristics of its elements at equilibrium, define sample volume and determine its size and shape. When water is removed from the material, a pressure unbalance is produced between the inner of the material and the external pressure. This generates contracting stresses that lead to material shrinkage or collapse, changing in shape and occasionally cracking of the product. Shrinkage of food materials increases with the volume of water removed, since the more the water removed the more contraction stresses are originated in the material (Adiletta et al., 2014). In some cases, changes in the material volume are equal to the volume of removed water (Krokida and Maroulis, 1997).

Macroscopic shrinkage is commonly referred to the ratio of the volume of the product at a given drying time to the initial volume. Based on the mass conservation and the volume additivity of the various product phases (dry solid, water and air), the change of the product volume as a function of moisture content is given as (De Lima et al., 2002):

$$\frac{V}{V_0} = 1 - a(X_0 - X) \quad (12)$$

where a is the shrinkage coefficient.

2.4. Porosity

Porosity is especially important in the reconstitution of the dried products, effectively controlling the speed of rewetting as well as taste and appearance. Information on pore formation and their characteristics in food during processing is also essential in estimating transport properties (i.e. thermal conductivity and diffusivity, mass diffusivity) and characterizing the quality of a dried product (Rahman, 2001). The porosity (ε) is the ratio between the volume of air present in the product and the overall volume and is expressed as a function of bulk and real density of product during drying process by Liu et al. (2012):

$$\varepsilon = 1 - \frac{\rho_b}{\rho_p} \quad (13)$$

The bulk density of the product is given as (Mercier et al., 2011):

$$\rho_b = \frac{1 + X}{\frac{1}{\rho_{b0}} + \frac{X}{\rho_w}} \quad (14)$$

where ρ_{b0} is the bulk density of dry product.

2.5. Effective moisture diffusivity

The moisture transfer in heterogeneous media can be conveniently analysed by using Fick's second law for homogenous materials, in which the heterogeneity of the material is taken into account by the use of effective moisture diffusivity. Fick's second law of diffusion has been frequently used to describe the internal moisture transfer, particularly for biological products during the falling drying rate period (Koua et al., 2009; Malafrente et al., 2015; Bezerra et al., 2015). This drying period is characterized by an internal mass transfer type drying with moisture diffusion as the controlling phenomenon.

The cocoa bean shape is assumed to be spherical and the Fick's second law of diffusion for spherical geometry is defined as follows (Hii et al., 2009a,b):

$$\frac{\partial X_r}{\partial t} = D_{eff} \left(\frac{\partial^2 X_r}{\partial r^2} + \frac{2}{r} \frac{\partial X_r}{\partial r} \right) \quad (15)$$

where r represents the radial coordinate, D_{eff} is the effective moisture diffusivity and X_r is the moisture ratio and defined as follows:

$$X_r = \frac{X - X_{eq}}{X_0 - X_{eq}} \quad (16)$$

where X_0 and X_{eq} represent the initial and equilibrium moisture content, respectively. X is the moisture content at given time.

By using appropriate initial and boundary conditions, Crank (1975) gave the analytical solution for spherical geometry as follows:

$$X_r = \frac{6}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \exp \left[-D_{eff} \left(\frac{n\pi}{R_p} \right)^2 t \right] \quad (17)$$

Initial and boundary conditions are as follows:

$$\begin{aligned} X_r &= X_0, \quad 0 \leq r \leq R \text{ at } t = 0 \\ X_r &= X_{eq}, \quad r = R \text{ at } t > 0 \\ \frac{\partial X_r}{\partial r} &= 0, \quad r = 0 \text{ at } t > 0 \end{aligned}$$

where R_p is the radius of cocoa bean.

Only the first term of Eq. (17) can be used to estimate the effective moisture diffusivity for long drying times:

$$X_r \cong \frac{6}{\pi^2} \exp \left(-D_{eff} \left(\frac{\pi}{R_p} \right)^2 t \right) \quad (18)$$

By plotting $\ln(X_r)$ versus time, the slope of the line will be constant of the above linear equation. Thus, the effective moisture diffusivity can be calculated using the following equation:

$$D_{eff} = \text{slope} \frac{R_p^2}{\pi^2} \quad (19)$$

2.6. Mass transfer

Moisture diffusion is the main mass transfer mechanism in the solid phase from the center of a product to the surface followed by convective moisture transfer to the air. The geometry considered is one-dimensional in the spherical coordinate system and the mass transfer is considered only in the radial direction. In the one-dimensional spherical coordinate system ($r \in [0, R]$), Fick's diffusion model of moisture transfer which is applied to simulate the time evolution of the distribution of the local moisture content of product during drying is as (Perussello et al., 2012):

$$\frac{\partial X}{\partial t} = \frac{D_{eff}}{r^2} \left[\frac{\partial}{\partial r} \left(r^2 \frac{\partial X}{\partial r} \right) \right] \quad (20)$$

The moisture content-depending boundary conditions at the surface and in the center-line take into account the mass convection to the surrounding air is given as follows:

$$\frac{\partial X}{\partial r} \Big|_{r=0} = 0 \quad (21)$$

$$-D_{eff} \frac{\partial X}{\partial r} \Big|_{r=R} = h_m (X - X_{eq}) \quad (22)$$

The initial condition is $X(r, t) = X_0$ at $t = 0$.

The mass transfer coefficient (h_m) in the surface of the cocoa beans was obtained according to the procedure described by Kaya et al. (2007) and Lemus-Mondaca et al. (2013). The boundary condition at the interface air/product can be written:

$$F_m = h_m \rho_d (X - X_{eq}) = -\frac{\rho_d}{\beta} \frac{dX}{dt} \quad (23)$$

After integration of Eq. (23), one may receive Eq. (24):

$$h_m = \frac{V}{At} \ln \left(\frac{X_0 - X_{eq}}{X - X_{eq}} \right) \quad (24)$$

Assuming that the cocoa bean shape is spherical, the mass transfer coefficient as a function of the drying curve can be calculated by the following formula:

$$h_m = \frac{d_p}{6t} \ln \left(\frac{X_0 - X_{eq}}{X - X_{eq}} \right) \quad (25)$$

2.7. Heat transfer

Heat transfer mechanisms include the internal heat generation and convective heat transfer mechanisms from/to the surface of the product to/from the drying air as well as evaporative cooling. In the one-dimensional spherical coordinate system ($r \in [0, R]$), the heat transfer is given by the following equation based on the Fourier's heat conduction model (Perussello et al., 2012):

$$\frac{\partial T}{\partial t} = \frac{\alpha}{r^2} \left[\frac{\partial}{\partial r} \left(r^2 \frac{\partial T}{\partial r} \right) \right] \quad (26)$$

The boundary conditions are symmetry and convective condition on the product centre and surface, respectively as it follows (Lemus-Mondaca et al., 2013):

$$\left. \frac{\partial T}{\partial r} \right|_{r=0} = 0 \quad (27)$$

$$-k \left. \frac{\partial T}{\partial r} \right|_{r=R} = h(T_p - T_a) + L_v h_m \rho_p (X - X_{eq}) \quad (28)$$

The initial condition is $T(r, t) = T_0$ at $t = 0$.

The evaporation at the product surface allows the coupling between the heat and mass transfer unsteady diffusion equation. The latent heat of vaporization (L_v) is given as (Youcef-Ali et al., 2001):

$$L_v = 4186.8(597 - 0.56(T + 273.15)) \quad (29)$$

For simulating the heat transfer, the Nusselt number (Nu) is derived from the Sherwood number (Sh) using the Lewis analogy equation (Incropera et al., 2013; Brenn et al., 2016):

$$\frac{Sh}{Nu} = \left(\frac{\alpha_a}{D_a} \right)^{\frac{1}{3}} = Le^{\frac{1}{3}} \quad (30)$$

The Nusselt number and the Sherwood number were obtained by the following correlations:

$$Nu = \frac{h d_p}{\lambda_a} \quad (31)$$

$$Sh = \frac{h_m d_p}{D_a} \quad (32)$$

By combining the Eqs. (30)–(32), the heat transfer coefficient (h) can be estimated by the following formula:

$$h = \frac{\rho_a C_{pa} d_p}{6t} Le^{\frac{2}{3}} \ln \left(\frac{X_0 - X_{eq}}{X - X_{eq}} \right) \quad (33)$$

where ρ_a is the air density, C_{pa} is the air specific heat capacity, α_a is the air thermal diffusivity and D_a is the water mass diffusivity in the air. The Lewis number ($Le = Sc/Pr$) has to be unity. For air, the Prandtl (Pr) and Schmidt (Sc) numbers are however not equal. $Pr \approx 0.71$ and $Sc \approx 0.6$, resulting in $Le \approx 0.85$ (Defraeye et al., 2012; Lecoq et al., 2016).

3. Materials and methods

3.1. Sample preparation

The cocoa beans used in this study were purchased from the local retail market. Fresh cocoa beans were retrieved from the fruits and were fermented inside a wooden box for five days. The beans were manually turned every 48 h to ensure uniformity during fermentation (Hii and Tukimon, 2002).

3.2. Indirect solar dryer

The indirect solar dryer has been designed and constructed at the Laboratory of New and Renewable Energy at the Polytechnic Institute of Yamoussoukro, Côte d'Ivoire. The indirect solar dryer (Fig. 1) consists mainly of a solar air collector system and a drying chamber containing three rectangular trays of 95 cm × 106 cm dimensions and a flux diffuser of 97 cm × 106 cm dimensions. Each tray is constructed of wooden frames on which is fixed nylon net to facilitate the air flow. The trays are separated of 15 cm to each other. The cocoa beans are laid out inside the drying chamber on the trays for drying. The solar air collector system, of 89 cm × 225 cm dimensions, was used to produce thermal energy for drying. The solar air collector consists of glass cover and aluminium absorber plate painted in black matte. It is tilted to an angle of 7° with respect to horizontal surface.

The basic operation of the indirect solar dryer is the following one: exposed to solar radiations, the solar air collector converts into heat a portion of radiant energy received on its surface. The air crossing this solar collector then receives a portion of this energy by convective heat exchange. This air, with better drying properties, circulates from the solar air collector to the drying



Fig. 1. Picture and schematic view of indirect solar dryer.

chamber containing trays on which are laid the cocoa bean for drying. Two fans regulate the airflow.

3.3. Drying procedure

Experiments on the indirect solar drying of cocoa beans were carried out in June–August 2016 on the site of our laboratory located at 6°48'36" N latitude and 5°17'44" W longitude. About 15 kg of fermented cocoa beans were spread thinly in one bean layer on tree trays inside the drying chamber. Drying was terminated when the cocoa beans reached equilibrium moisture content by prolonging the drying process until no further change in weight was observed.

3.4. Moisture content

To evaluate cocoa bean dry mass, sample was taken on each tray at the end of drying. These samples were weighted with an electronic balance of 0.01gr precision and placed in an oven-dryer at 105 °C for 24 h. At the end of the 24 h of oven-drying, the samples were reweighted and the average mass for these samples was the dry mass of the cocoa beans. The beans moisture content at different stages of drying was determined with reference to the dry mass of the cocoa beans using Eq. (34):

$$X = \frac{m - m_d}{m_d} \quad (34)$$

where m and m_d are the mass at instant t and the dry mass, respectively, and X represents the cocoa beans moisture content in dry basis.

The equilibrium moisture content (X_{eq}) was determined by prolonging the drying process until no further change in weight was observed for the cocoa beans. The initial moisture content of cocoa beans was found to be 1.22 kg water/kg db and the X_{eq} value determined was 0.0698 kg water/kg db.

3.5. Desorption isotherms

Desorption isotherms of cocoa beans were determined using the static method, which is based on using saturated salt solutions to maintain constant water activity of samples once equilibrium between room conditions and food sample is reached. Salt solutions used to obtain constant relative humidity were LiCl, KCH₃CO₂, MgCl₂, K₂CO₃, Mg(NO₃)₂, NaBr, SrCl₂, NaCl, KCl and BaCl₂. This group of salts allows obtaining a wide range of relative humidity from 11% to 90% as reported by Kiranoudis et al. (1993), Prette et al. (2013) and Koua et al. (2014). Experimental data were fitted using the Guggenheim, Anderson and de Boer (GAB) model, frequently used in food studies. This model has three parameters that have physical meanings and the model has been found to adequately represent the water activity experimental data over the range of practical interest for most foods (Vigano et al., 2012; Koua et al., 2014).

$$X_{eq} = \frac{X_m G K a_w}{(1 - K a_w)(1 + (G - 1) K a_w)} \quad (35)$$

where X_m is the moisture content corresponding to the formation of a monomolecular layer on the internal surface; G is a constant related to the heat of sorption of the first layer on primary sites and K is a factor correcting properties of the multi-layer molecules with respect to the bulk liquid.

3.6. Thermophysical properties

Thermophysical properties of cocoa beans were estimated by using formulas existing in the literature food products. Specific

heat and thermal conductivity were obtained from Eqs. (36) and (37) which are function of the moisture content.

Specific heat (Huang et al., 2013):

$$C_p = 837 + 3348 \left(\frac{X}{1 + X} \right) \quad (36)$$

Thermal conductivity (Ruiz-Lopez et al., 2004):

$$k = 0.49 - 0.44 \exp(-0.206X) \quad (37)$$

The thermal diffusivity was calculated by the following:

$$\alpha = \frac{k}{\rho C_p} \quad (38)$$

3.7. Shrinkage

The shrinkage curves were established by measuring sample in all three orthogonal axes during drying. Measurements were carried out using a caliper of 0.05 mm precision. Shrinkage represents the volume reduction or the change of selected dimensions occurring due to the moisture taken away from the cocoa beans structure in a drying process. The shrinkage (S) is calculated as follows:

$$S = 1 - \frac{V}{V_0} \quad (39)$$

where V is the volume of cocoa bean at any time and V_0 is its initial volume.

$$V_0 = \frac{1}{6} \pi d_{p0}^3 \text{ and } V = \frac{1}{6} \pi d_p^3 \quad (40)$$

Using of geometric mean diameter is necessary for the calculation of the volume because cocoa bean has a spheroid shape as illustrated in Fig.2. The geometric diameter (d_p) of fresh and dried cocoa beans can be calculated as follows during the drying process (Delgado et al., 2014; Chayjan and Kaveh, 2014):

$$d_p = (x \cdot y \cdot z)^{1/3} \quad (41)$$

where x , y and z are major, intermediate and minor diameters of the bean, respectively. Characteristic dimensions of category B cocoa beans were determined as a function of bean moisture content in the moisture range between 1.22 kg water/kg db and 0.074 kg water/kg db. Thereby, $2.5 \text{ cm} \leq x \leq 2.8 \text{ cm}$, $1.3 \text{ cm} \leq y \leq 1.6 \text{ cm}$ and $0.9 \text{ cm} \leq z \leq 1.2 \text{ cm}$.

4. Results and discussion

4.1. Composition and thermophysical properties of cocoa beans

Composition data for cocoa beans are readily available by Ohene Afoakwa et al. (2013). These data consist of the centesimal

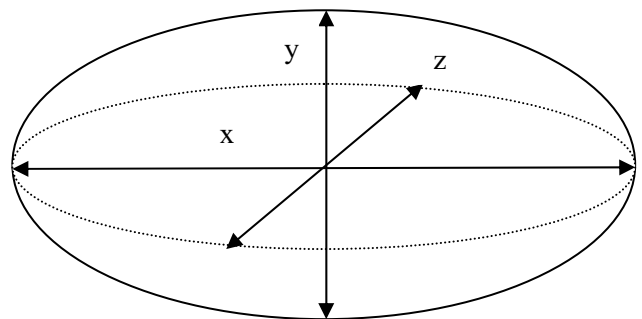


Fig. 2. Schematic drawing of a cocoa bean with its characteristic dimensions.

Table 1

Centesimal composition of dried cocoa beans (Ohene Afoakwa et al., 2013).

Component	% Wet basis
Water	4 ± 0.02
Carbohydrates	21 ± 0.08
Fats	53.4 ± 0.63
Proteins	18.8 ± 0.56
Ashes	2.8 ± 0.07

Table 2

Cocoa bean thermophysical properties in function of moisture content in dry basis.

Properties	Equations	r^2
Density (kg/m ³)	$\rho_p = 147.95 \frac{x}{x_0} + 691.46$	0.978
Specific heat (J/kg K)	$C_p = 2623.35 \left(\frac{x}{x_0}\right)^{0.33}$	0.995
Thermal conductivity (W/m K)	$k = 0.0987 \frac{x}{x_0} + 0.0513$	0.999
Thermal diffusivity (m ² /s)	$\alpha = 7.37 \times 10^{-8} \left(\frac{x}{x_0}\right)^{-0.11}$	0.935

composition of the major components found in food items. Such components include water, proteins, fats, carbohydrates and ashes as presented in Table 1.

The thermophysical properties of cocoa beans are described by the equations presented in Table 2, which were fitted to the experimental data. Cocoa bean thermophysical properties were estimated as function of moisture content as reported by Perussello et al. (2014) and Tzempelikos et al. (2015).

The characterization of cocoa beans in spherical coordinates was made with the sizes during drying. Cocoa bean characteristics dimensions were estimated from the major, intermediate and minor diameters. The average values obtained are listed in Table 3. The cocoa beans used in this study are of category B according to the quality control division of the Ghana cocoa board (Bart-Plange et al., 2012). Cocoa beans are sorted by size based on a number of beans weighing 100 g into various categories. Table 4 shows five categories that are declared for the main harvesting season.

4.2. Desorption isotherms

Fig. 3 shows the experimental desorption isotherms for cocoa beans at 30, 45 and 60 °C with its respective fitting to the GAB model. Desorption isotherm shows an increase in equilibrium moisture content with increasing water activity, at constant temperature. The equilibrium moisture content decreases with the temperature at constant water activity. The GAB model was tested for adequacy and goodness of fit by determining the coefficient of correlation (r^2), mean relative percentage deviation modulus (E), root mean square error (RMSE) and residual plots. These values and the model parameters obtained by nonlinear regression analysis are presented in Table 5. This model presented a good fit for the experimental data of the cocoa beans. The GAB model is the most widely accepted and representative model for sorption isotherms of foods (ASAE, 2004), mainly due to its accuracy and validity over a wide range of water activities from 0.11 to 0.90.

Table 3

Cocoa bean characteristics dimensions.

	Sample mass (g)	Moisture content (kg water/kg db)	Major x (cm)	Intermediate y (cm)	Minor z (cm)	Geometric diameter (cm)
Initial	2.4	1.22	2.76	1.55	1.13	1.69
Final	1.16	0.074	2.58	1.38	0.93	1.49

Table 4

Categories of cocoa beans.

Classification	Beans count/100 g
Large (category B)	0–100
Medium (category D)	101–120
Small (category G)	121–130
Very small (category F)	131–150
Remnants (category SR)	151–250

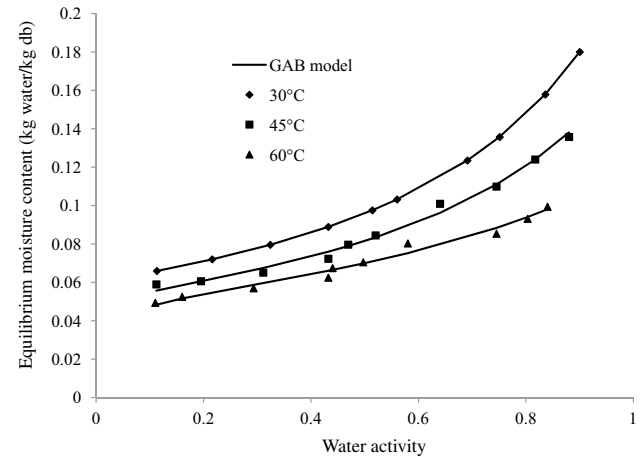


Fig. 3. Experimental desorption isotherms of cocoa beans at 30, 45 and 60 °C and estimated curves with GAB model.

4.3. Drying kinetics

The moisture content of the cocoa beans as a function of drying time is presented in Fig. 4 for convective drying in an indirect solar dryer. The cocoa bean drying curve can be divided into two parts as illustrated in this Fig. 4. Similar results have been reported for drying of cocoa beans (Hii et al., 2009a,b), mango slices and milky mushroom (Arumuganathan et al., 2009), and jackfruits (Kaushal and Sharma, 2016). The moisture content rapidly reduces and then slowly decreases as drying progresses. Initially, a rapid decrease in the drying rate occurs until a moisture content value is close to 0.23 kg water/kg db. After this value of moisture content, the drying rate decreases slowly to reach equilibrium moisture content. During this second falling drying rate period, the cocoa beans are in the hygroscopic domain. In this domain, water exits only in tied form by sorption in the product and in vapour form. Therefore, the characteristic dimensions of cocoa beans decrease very slowly. As it follows from the drying curve in Fig. 4, indirect solar drying is a very long process, as it takes about 20 h, on average.

The moisture ratio of cocoa beans as a function of drying time is presented in Fig. 5. The moisture ratio of cocoa beans decreased exponentially as the drying time increased during the drying process. Continuous decrease in moisture ratio indicates that diffusion has governed the internal mass transfer.

It can be seen from Fig. 6 that the mass flux decreased continuously throughout the drying period, which indicated that cocoa bean drying process took place in the falling rate period. It can be considered as a diffusion controlled process that the rate of

Table 5
Parameter estimated values, mean relative percentage deviation modulus (E), root mean square error (RMSE), correlation coefficient (r^2) and residual plots for GAB model to cocoa bean desorption process.

Temperature ($^{\circ}\text{C}$)	Parameters			r^2	E (%)	RMSE	Residual plots
	X_m	G	K				
30	0.0606	3729.9748	0.7372	0.999	0.0392	0.00006	Random
45	0.0535	279.5272	0.6970	0.990	2.4083	0.0025	Random
60	0.0508	115.6945	0.5789	0.978	2.7397	0.0024	Random

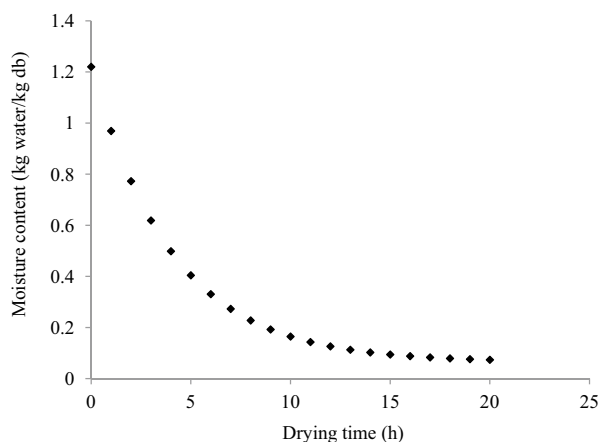


Fig. 4. Variation in moisture content of cocoa beans during indirect solar drying.

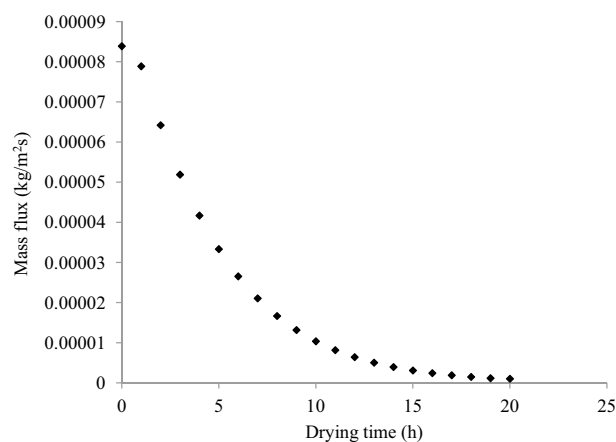


Fig. 6. Variation in mass flux of cocoa beans during indirect solar drying.

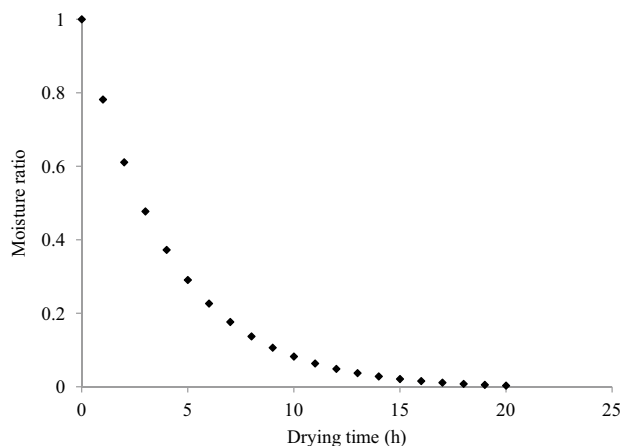


Fig. 5. Variation in moisture ratio of cocoa beans during indirect solar drying.

removed moisture is limited by diffusion of moisture from inside to surface (Duc et al., 2011). Comparable results have also been reported for various other agricultural products (Hii et al., 2009a, b; Liu et al., 2012). The experimental mass flux curve was tested by several models. Among these models, that of relation (42) may be assumed to fit the most suitably cocoa bean drying rate profile.

$$F_m = 0.317(X_0 - X_{eq}) \exp(-0.23t) \quad (r^2 = 0.998) \quad (42)$$

where F_m is in $\text{kg/m}^2 \text{ h}$ and t is in hour (h).

4.4. Effective moisture diffusivity

By using Eq. (19), curve of effective moisture diffusivity versus reduced moisture content is presented in Fig. 7. It can be found that the effective moisture diffusivity decreases with the decrease in reduced moisture content as reported by Silva et al. (2016). Additionally, as reported in many drying literatures, the effective

moisture diffusivity increased with drying temperature. Torrez Irigoyen et al. (2014) reported that the effective diffusion coefficient was assessed as function of the product temperature and moisture content. This fact could be explained as increase in drying temperature causes a higher driving force speeding up transfer of water vapour from the core to the surface of cocoa beans resulting in high diffusivity (Falade and Oyedele, 2010).

The values of effective moisture diffusivities were found to in range from 5.49×10^{-10} to $4.26 \times 10^{-10} \text{ m}^2/\text{s}$. These values are within the range of diffusivities reported for most agricultural crop materials (Panagiotou et al., 2004). The effective moisture diffusivity of cocoa beans may be represented by the following relationship with the corresponding moisture content:

$$D_{eff} = 1.30 \times 10^{-10} \frac{X}{X_0} + 4.18 \times 10^{-10} \quad r^2 = 0.999 \quad (43)$$

4.5. Shrinkage

The volume ratio and shrinkage coefficient versus reduced moisture content of cocoa beans are presented in Fig. 8. In each case, five beans were used and the average values are considered. As shown in this figure, the volume ratio decreases with decrease in reduced moisture content of cocoa beans. Similarly, shrinkage coefficient increases with decrease in reduced moisture content of cocoa beans. Shrinkage curves were almost linear in the cocoa bean drying process as reported by Silva et al. (2016). Fig. 8 shows that the moisture content has significant effect on the shrinkage curves. This result is also predictable and can be due to the slow drying rate at low temperatures and moisture distribution which is uniform in the cocoa beans. This case reduces internal stresses and leads to continuous shrinkage until the end of drying process (Horuz and Maskan, 2015).

Fig. 9 shows the result of comparison curve between the shrinkage experimental and fitted data. The estimated value of shrinkage coefficient in Eq. (12) applied to cocoa beans was $a = 0.276$ with a

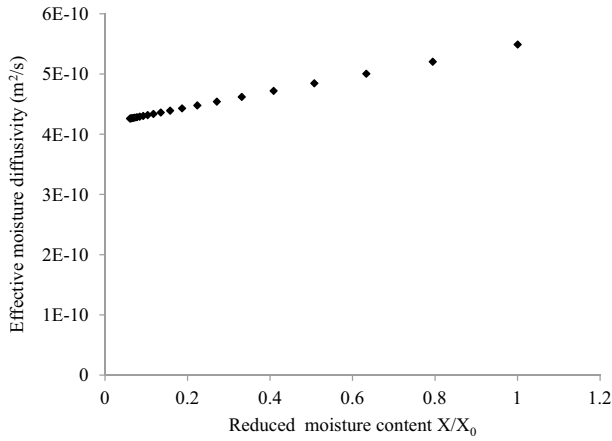


Fig. 7. Effect of moisture content variations on effective moisture diffusivity of cocoa beans.

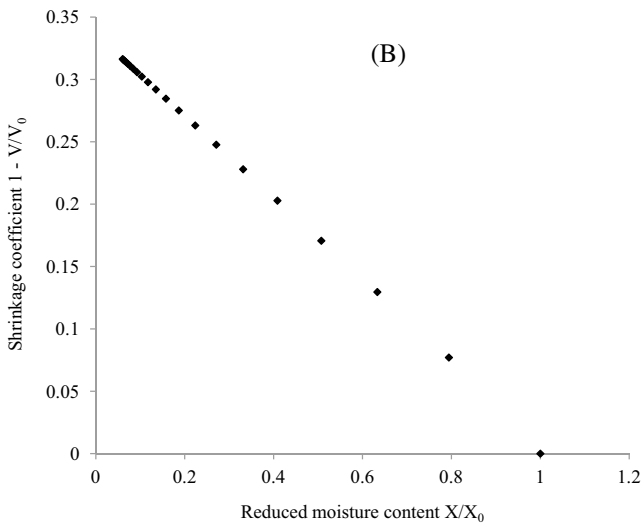
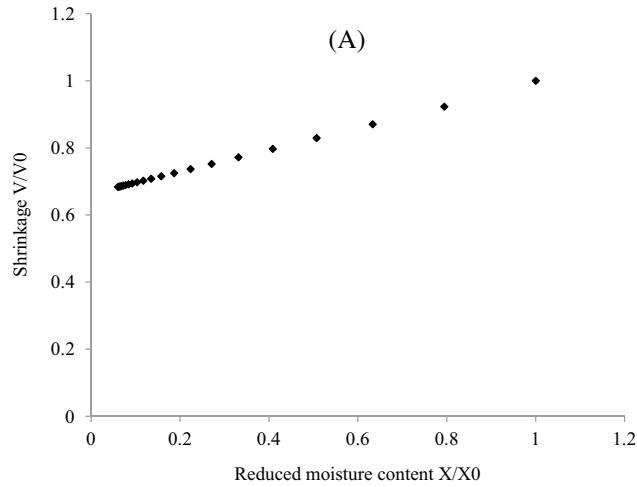


Fig. 8. Effect of moisture content variations on volume ratio (A) and shrinkage coefficient and (B) of cocoa beans.

correlation coefficient of $r^2 = 0.999$. This shrinkage coefficient indicates that the cocoa bean volume is decreased about 0.276 mm^3 per each mm^3 of water removed.

By rearranging Eq. (12), shrinkage can be expressed in terms of the bean diameter (Eq. (44)). This expression can be used to predict

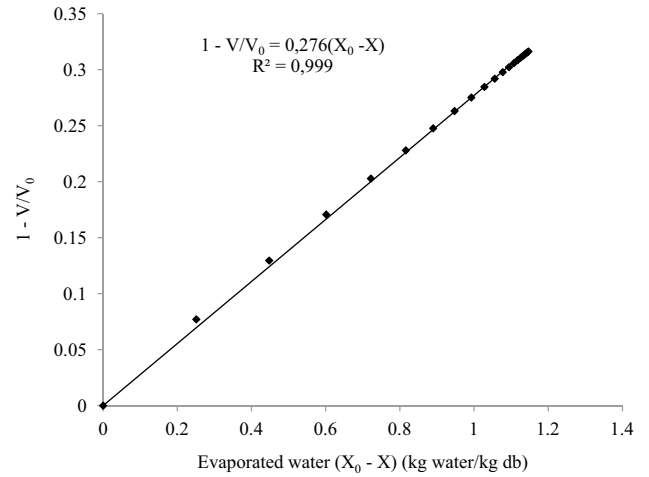


Fig. 9. Shrinkage coefficient versus evaporated water during indirect solar drying of cocoa beans.

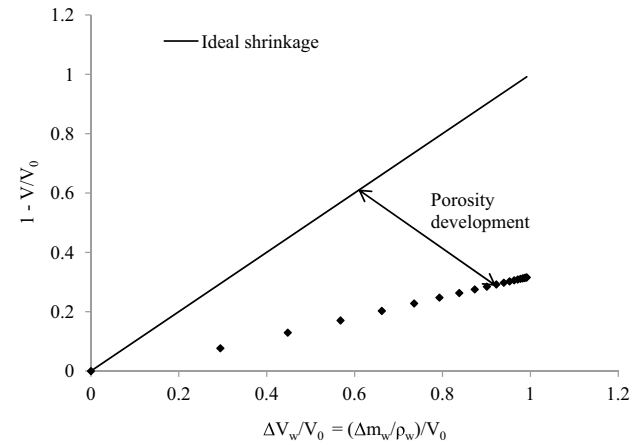


Fig. 10. Cocoa beans volume reduction versus the volume of water removed during indirect solar drying.

the variation of the cocoa bean diameter as a function of moisture content and thereby with time:

$$\frac{d_p}{d_0} = (1 - 0.276(X_0 - X))^{\frac{1}{3}} \quad (44)$$

Fig. 10 shows the cocoa bean volume reduction versus the volume of water removed. On this figure, significant deviations were observed between the curves of cocoa bean volume reduction and the bisector representing a situation of ideal shrinkage where the water removed is converted exactly to a loss of volume (Dissa et al., 2010). The volume of water removed was greater than the cocoa bean volume reduction during all drying. During drying, the cocoa bean undergoes change from rubbery to glassy state (Kurozawa et al., 2012). The rigidity of the material causes a decrease in the rate volume change and parallel pore formation may happen (Kurozawa et al., 2012). As shown in Fig. 10, this deviation represented the porosity development of cocoa beans during drying. Most of the water removed from the cocoa bean was thus replaced by gas (Dissa et al., 2010). The dried cocoa bean is therefore a porous product made up of gas and dry matter.

4.6. Real density

Fig. 11 shows the variations of the real density with reduced moisture content during the cocoa beans drying. The real density

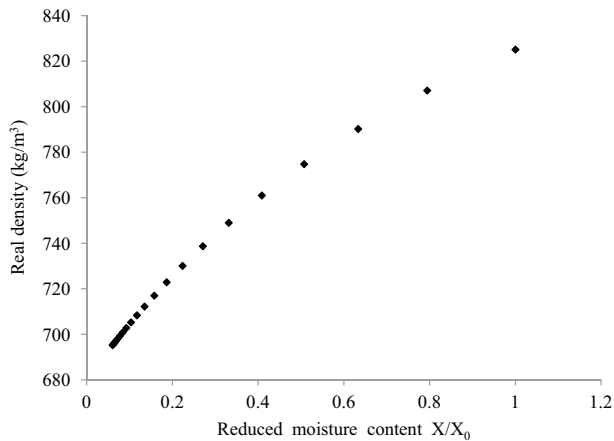


Fig. 11. Effect of moisture content variations on real density of cocoa beans.

decreases as reduced moisture content decreased, from an average value of 825.10 kg/m^3 at a moisture content of $1.22 \text{ kg water/kg db}$ to 695.25 kg/m^3 at a moisture content of $0.074 \text{ kg water/kg db}$. This behaviour was not affected by drying method (Koç et al., 2008). The variation may be due to a higher decrease in mass of the cocoa bean as compared to its volumetric decrease as cocoa bean moisture content decreases. A similar behaviour was also reported by other authors for fruits and vegetables with various geometrics (Bart-Plange and Baryeh, 2003; Hatamipour and Mowla, 2003). The real density was fitted by the following formula:

$$\rho_p = 147.95 \frac{X}{X_0} + 691.46 \quad (r^2 = 0.978) \quad (45)$$

4.7. Porosity

Fig. 12 shows the variation of porosity with reduced moisture content during the cocoa bean drying. The porosity increased as reduced moisture content decreased, from an average value of 15.82% at a moisture content of $1.22 \text{ kg water/kg db}$ to 24.67% at a moisture content of $0.074 \text{ kg water/kg db}$. Porosity in fruits and vegetables increases during drying process depending on the initial moisture content, composition and size, as well as on the type of drying (Koç et al., 2008). During drying, the pores previously occupied by water either are replaced by air or is compressed as result

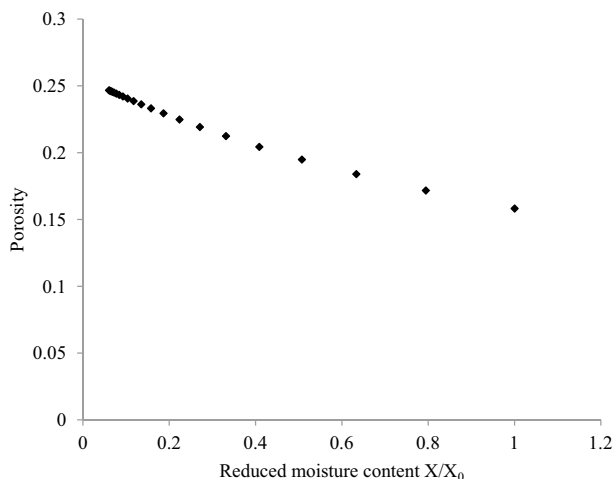


Fig. 12. Effect of moisture content variations on porosity of cocoa beans.

of shrinkage (Zabalaga et al., 2016). The relationship between porosity and reduced moisture content is given by the following:

$$\varepsilon = -0.099 \frac{X}{X_0} + 0.25 \quad (r^2 = 0.985) \quad (46)$$

4.8. Heat and mass transfer coefficients

Heat and mass transfer occur simultaneously during the drying process. Fig. 13 shows the variation of the heat and mass transfer coefficients with the drying time. As presented in Fig. 13, the heat and mass transfer coefficients increased with the increasing drying time. During the indirect solar drying, the increase in drying time increases the drying air temperature. The increase in drying air temperature increases the temperature of the food material. But, the increase in temperature of the moist food material increases the moisture diffusivity, which in turn increases the rate of diffusion of moisture from inside to the surface of the food material. It means that mass transfer coefficient increases with increasing drying air temperature. Similarly, the heat transfer coefficient increases with increasing drying air temperature because the heat transfer coefficient is calculated using the heat and mass transfer analogy. This behaviour is in agreement with literature data on heat and mass transfer (Ateeque et al., 2014; Fu et al., 2015). The values of h and h_m were in the range from 1.92×10^{-4} to $8.08 \times 10^{-2} \text{ W/m}^2 \text{ K}$ and from 1.88×10^{-7} to $7.88 \times 10^{-5} \text{ m/s}$, respectively, for a convective drying of cocoa beans. These results were found to be in the range of those available in the literature for different fruits and vegetables and drying conditions (Lemus-

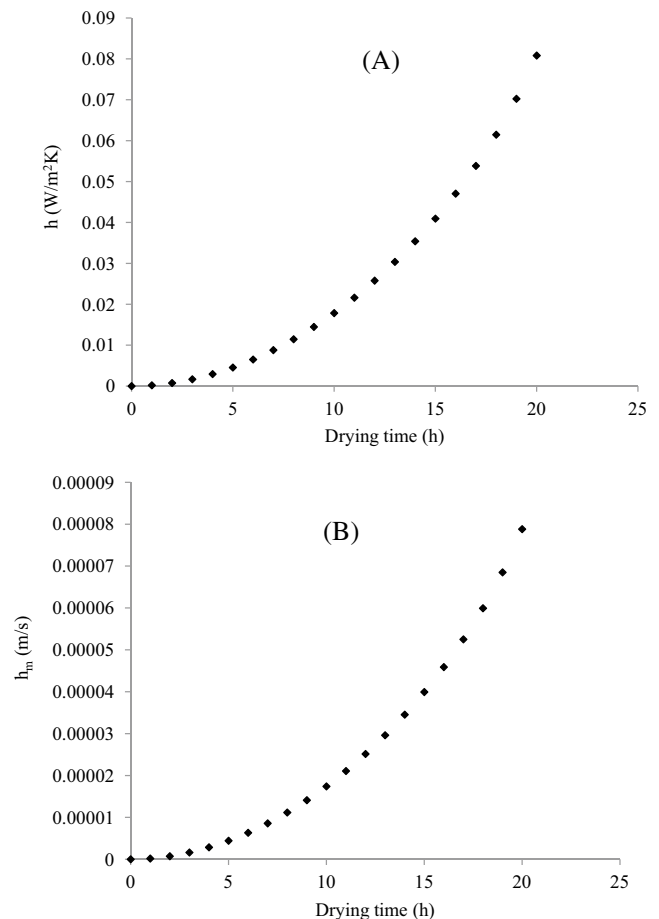


Fig. 13. Variation in heat (A) and mass (B) transfer coefficients during indirect solar drying.

Mondaca et al., 2013). Heat and mass transfer coefficients of cocoa beans are dependent on moisture content or temperature and are also influenced by structural changes occurring to cocoa beans during drying. The heat and mass transfer coefficients were allowed to change with drying time in the polynomial form:

$$h = 2.21 \times 10^{-4} t^2 - 5.83 \times 10^{-4} t + 1.20 \times 10^{-3} \quad (r^2 = 0.998) \quad (47)$$

$$h_m = 7.75 \times 10^{-4} t^2 - 2.05 \times 10^{-4} t + 4.23 \times 10^{-3} \quad (r^2 = 0.998) \quad (48)$$

5. Conclusion

In this work, the evolution of shrinkage, density, porosity, heat and mass transfer coefficients was studied. The following conclusions are drawn from this investigation:

- The cocoa bean thermophysical properties were obtained and fitted to nonlinear correlations, describing their behaviour as a function of moisture content.
- The values of effective moisture content were calculated by Fick's diffusion model and their values varied from 5.49×10^{-10} to 4.26×10^{-10} m²/s. D_{eff} decreased with decrease in moisture content.
- The shrinkage decreased linearly with decrease in moisture content.
- The real density decreased from 825.10 to 695.25 kg/m³ with decrease in moisture content.
- The porosity increased from 15.82 to 24.67% with decrease in moisture content.
- The heat and mass transfer coefficients increased during the drying process. The values of h and h_m were in the range from 1.92×10^{-4} to 8.08×10^{-2} W/m² K and from 1.88×10^{-7} to 7.88×10^{-5} m/s, respectively.

Author's contributions

All authors of the paper contributed equally to the writing of the paper and were involved in the overall planning and supervision of the work.

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